

UNIVERSITY OF CLAUDE · COMPLETE CATALOG MAP

Free-Resource Catalog

Every course in the CSCE graduate catalog, mapped to open courseware, free books, and public lecture recordings

HOW TO READ THIS

Each entry gives the course, a one-line statement of what the subject actually is, and the best freely available ways to learn it.

Every link is free: open courseware, an author-released book, a public lecture recording, or a free-audit MOOC. Nothing here requires payment. Resources remain under their own licences.

84 courses · 268 resources. Courses in the Graphics & Game Engines degree plan are marked ■.

Course	What it is, and where to learn it free
CSCE 706 Structured Programming in C	<i>C from first principles: memory, pointers, and the machine model under every other language.</i> · Harvard CS50 — full course, free · Modern C (Jens Gustedt, free book) · Beej's Guide to C Programming
CSCE 708 Intro to CS Concepts and Programming (Java)	<i>Algorithmic thinking and iterative problem solving, taught through Java.</i> · MOOC.fi Java Programming I & II (free, graded) · Harvard CS50 · MIT 6.0001 Intro to CS (OCW)
CSCE 709 Intermediate Program Design with C++	<i>Object-oriented and value-semantic C++: RAII, dynamic memory without leaks, error handling.</i> · learncpp.com — complete modern C++ course · cppreference · CppCon talks (free, YouTube)
UC MATH 600 Mathematics for Computer Science (program prerequisite)	<i>Proof technique, combinatorics, graphs, and discrete probability — the language every later course assumes.</i> · MIT 6.1200J / 6.042J Mathematics for Computer Science (OCW, full video) · MIT 6.1200J current lecture series · Lehman, Leighton & Meyer, Mathematics for CS (free book)

Course	What it is, and where to learn it free
UC LINA 600 Linear Algebra for Graphics (program prerequisite)	<i>Vectors, matrices, bases, eigen-decomposition, and projections — the entire vocabulary of the graphics pipeline.</i> · MIT 18.06 Linear Algebra — Gilbert Strang (OCW, full video) · 3Blue1Brown, Essence of Linear Algebra (free video) · Immersive Math — interactive linear algebra (free)
CSCE 603 Database Systems and Applications	<i>Codd's relational model and hands-on schema design for non-majors.</i> · Stanford Databases MOOC (Jennifer Widom, free) · CMU 15-445 Intro to Database Systems (free videos) · Stanford CS145 Introduction to Databases (archived) · PostgreSQL Tutorial
■ CSCE 608 Database Systems	<i>How a DBMS is actually built: storage, indexing, query optimization, concurrency, recovery.</i> · CMU 15-445/645 — Andy Pavlo, full lectures free · CMU 15-721 Advanced Database Systems · MIT 6.5830 Database Systems, Fall 2023 (OCW) · Database Internals / readings list
CSCE 605 Compiler Design	<i>Beyond parsing: dataflow analysis, SSA, register allocation, and optimization passes.</i> · Stanford CS143 Compilers (free, edX/self-paced) · Crafting Interpreters (Bob Nystrom, free book) · Neso Academy, Compiler Design (free video series) · LLVM Kaleidoscope tutorial
■ CSCE 606 Software Engineering	<i>Building software that survives other people: process, design, testing, maintainability.</i> · Berkeley CS169 — Engineering Software as a Service (edX, free audit) · MIT 6.031 Software Construction (open) · Software Engineering at Google (free online)
■ CSCE 611 Operating Systems	<i>Scheduling, virtual memory, concurrency, and I/O — by building a kernel, not reading about one.</i> · MIT 6.1810 Operating System Engineering (xv6, free) · OSTEP — Operating Systems: Three Easy Pieces (free book) · MIT 6.S081 Operating System Engineering, 2020 (lecture video + labs) · Berkeley CS162 (free lectures)
CSCE 612 Applied Networks and Distributed Processing	<i>The layered stack as engineering practice: build a working TCP, then reason about real protocols.</i> · Stanford CS144 — Computer Networking (free, build your own TCP) · Kurose & Ross companion site · Beej's Guide to Network Programming

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■ CSCE 614 Computer Architecture	<i>Pipelines, caches, memory hierarchy, multicore, and why your code is slow.</i> · ETH Zurich — Onur Mutlu, Computer Architecture (full lectures) · Nand2Tetris — build a computer from NAND gates · MIT 6.004 Computation Structures (OCW, Spring 2017) · MITx 6.004.1x Computation Structures: Digital Circuits (edX) · Berkeley CS152/CS252
CSCE 616 Intro to Hardware Design Verification	<i>Proving a chip does what the spec says: testbenches, coverage, assertions.</i> · ChipVerify — SystemVerilog & UVM tutorials (free) · Verilator (open-source simulator) docs · Verification Academy (free account)
CSCE 714 Advanced Hardware Design Functional Verification	<i>UVM at scale: reusable verification IP, constrained-random stimulus, coverage closure.</i> · ChipVerify UVM tutorial · Accellera UVM standard (free) · cocotb — Python verification (free)
CSCE 620 Computational Geometry	<i>Convex hulls, Voronoi diagrams, triangulation, range search — the geometry under collision and meshing.</i> · David Mount, CMSC 754 lecture notes (free PDF) · CGAL — the reference implementation · Computational Geometry (TU Eindhoven course page)
CSCE 624 Sketch Recognition	<i>Turning freehand strokes into meaning: feature-, geometry-, and timing-based recognizers.</i> · \$1 / \$N / \$P stroke recognizer family (free code + papers) · Berkeley CS188 (classification foundations) · Sketch-a-Net / QuickDraw dataset
CSCE 625 Artificial Intelligence	<i>Search, constraint satisfaction, adversarial games, logic, planning, probabilistic reasoning.</i> · Berkeley CS188 — Intro to AI (free, with Pacman projects) · AIMA book companion site · MIT 6.034 — Patrick Winston lectures (OCW) · Harvard CS50's Introduction to AI with Python (free)
CSCE 626 Parallel Algorithm Design and Analysis	<i>Work-span analysis, parallel lower bounds, and algorithms built for wide machines.</i> · CMU 15-210 Parallel and Sequential Data Structures (free notes) · Blelloch, Prefix Sums and Their Applications (free) · CMU 15-418 Parallel Computer Architecture
CSCE 627 Theory of Computability	<i>Automata, Turing machines, decidability, and the hard edge of what computers cannot do.</i> · MIT 18.404J Theory of Computation — Sipser, full video (OCW) · Stanford CS103 handouts (free) · Jeff Erickson, Models of Computation (free book)

Course	What it is, and where to learn it free
CSCE 637 Complexity Theory	<i>P, NP, space, randomness, alternation, and the relationships between complexity classes.</i> · Arora & Barak, Computational Complexity (free draft) · CMU 15-455 Undergraduate Complexity — Ryan O'Donnell (free video) · Computational Complexity Blog
CSCE 628 Computational Biology	<i>Genomics and proteomics restated as string, graph, and optimization problems.</i> · MIT 6.047 Computational Biology — Kellis (OCW + video) · Rosalind — bioinformatics problem set (free) · UCSD Bioinformatics Specialization (free audit)
■ CSCE 629 Analysis of Algorithms	<i>Design techniques, rigorous analysis, NP-completeness, approximation and randomization.</i> · MIT 6.046J Design and Analysis of Algorithms (OCW, full video) · Jeff Erickson, Algorithms (free book + exercises) · MIT 6.006 Introduction to Algorithms, Spring 2020 (OCW, full video) · Tim Roughgarden, Algorithms Illuminated (free lectures)
CSCE 658 Randomized Algorithms	<i>Probabilistic method, concentration bounds, and algorithms that are fast because they gamble.</i> · Jeff Erickson, randomization chapters (free) · Harvard CS 225 Sketching Algorithms — Jelani Nelson (free video+notes) · Mitzenmacher & Upfal lecture slides (free)
CSCE 669 Computational Optimization	<i>Polyhedral combinatorics, LP/IP duality, and the geometry behind approximation guarantees.</i> · Boyd & Vandenberghe, Convex Optimization (free book + EE364A video) · MIT 15.093 Optimization Methods (OCW) · Stanford EE364A lecture videos
CSCE 717 Algorithmic Game Theory	<i>Equilibria, mechanism design, auctions, and the price of letting agents act selfishly.</i> · Tim Roughgarden, CS364A — Twenty Lectures on AGT (free notes + video) · Algorithmic Game Theory (Nisan et al., free PDF) · Shoham & Leyton-Brown, Multiagent Systems (free)
CSCE 640 Quantum Algorithms	<i>The circuit model, Deutsch–Jozsa through Shor and Grover, and what quantum speedup really means.</i> · IBM Quantum Learning — full free curriculum · John Watrous, Quantum Information lectures (free) · Berkeley CS191 / edX Quantum Mechanics & Quantum Computation
CSCE 633 Machine Learning	<i>Supervised and unsupervised learning from the ground up: bias/variance, kernels, ensembles.</i> · Stanford CS229 — full lectures + notes (free) · An Introduction to Statistical Learning (free PDF) · The Elements of Statistical Learning (free PDF)

Course	What it is, and where to learn it free
CSCE 636 Deep Learning	<i>Neural architectures, optimization, regularization, and modern representation learning.</i> · Karpathy, Neural Networks: Zero to Hero (free) · Dive into Deep Learning (free interactive book) · Stanford CS231n — CNNs for Visual Recognition · MIT 6.S191 Introduction to Deep Learning (free, annual) · fast.ai Practical Deep Learning
CSCE 642 Deep Reinforcement Learning	<i>MDPs, policy gradients, Q-learning, actor-critic, and model-based RL.</i> · Berkeley CS285 Deep RL — full lectures (free) · Sutton & Barto, Reinforcement Learning (free book) · OpenAI Spinning Up in Deep RL · David Silver, DeepMind RL course (free video)
CSCE 638 Natural Language Processing	<i>Language models, syntax, semantics, and the application stack from IE to dialogue.</i> · Stanford CS224N — NLP with Deep Learning (free video) · Jurafsky & Martin, Speech and Language Processing (free draft) · Hugging Face NLP Course (free)
CSCE 630 Speech Processing	<i>Acoustics, feature extraction, recognition, synthesis, and voice modification.</i> · Jurafsky & Martin SLP3 — speech chapters (free) · ESPnet end-to-end speech toolkit + tutorials · Kaldi documentation
CSCE 666 Pattern Analysis	<i>Density estimation, dimensionality reduction, clustering, Bayesian and geometric classifiers.</i> · Bishop, Pattern Recognition and Machine Learning (free PDF) · The Elements of Statistical Learning (free PDF) · scikit-learn user guide
CSCE 676 Data Mining and Analysis	<i>Pattern mining, clustering, classification, and analysis at dataset scale.</i> · Mining of Massive Datasets — free book + Stanford CS246 video · Zaki & Meira, Data Mining and Machine Learning (free PDF) · Stanford CS246
CSCE 727 Algorithmic Foundations of Big Data Computing	<i>Streaming, sketching, sublinear-time and sublinear-space algorithms.</i> · Sketching Algorithms — Jelani Nelson (free video + notes) · Mining of Massive Datasets · Data Streams: Algorithms and Applications (Muthukrishnan, free)
CSCE 726 Large-Scale Optimization for Machine Learning	<i>SGD and its descendants: momentum, adaptivity, variance reduction, convergence proofs.</i> · Bubeck, Convex Optimization: Algorithms and Complexity (free) · Boyd, Convex Optimization + EE364B · Optimization for Machine Learning (Hazan, free lecture notes)

Course	What it is, and where to learn it free
CSCE 631 Intelligent Agents	<i>Agent architectures, coordination, negotiation, and multi-agent equilibria.</i> · Shoham & Leyton-Brown, Multiagent Systems (free book) · Berkeley CS188 — multi-agent search · Roughgarden CS364A (mechanism design)
CSCE 635 AI Robotics	<i>Perception-action loops for mobile robots, grounded in biology and cognition rather than control theory.</i> · Modern Robotics — free book, video, Coursera · Peter Corke, Robot Academy (free video) · UC Berkeley CS287 Advanced Robotics — Pieter Abbeel (free) · ROS 2 tutorials
CSCE 752 Robotics and Spatial Intelligence	<i>Frames, kinematics, motion planning, calibration, SLAM, and navigation.</i> · Modern Robotics (Lynch & Park) — free book + lectures · LaValle, Planning Algorithms (free book) · Cyrill Stachniss, Mobile Sensing & Robotics (free video)
CSCE 753 Computer Vision and Robot Perception	<i>Projective geometry, multi-view reconstruction, features, lidar, and SLAM.</i> · Szeliski, Computer Vision: Algorithms and Applications (free book) · First Principles of Computer Vision — Shree Nayar (free video) · Stachniss SLAM course (free video)
CSCE 634 Intelligent User Interfaces	<i>Interfaces that model and adapt to their user, task, and context.</i> · Scott Klemmer, HCI (free audit) · Nielsen Norman Group articles (free) · Stanford CS 347 / HCI reading lists
■ CSCE 641 Computer Graphics	<i>The full pipeline: transforms, rasterization, shading, visibility, texturing, animation.</i> · GAMES101 — Lingqi Yan, Modern Computer Graphics (free, English slides) · CMU 15-462/662 — Keenan Crane (free video) · Cem Yuksel, Interactive Computer Graphics (Utah, free video) · UC San Diego CSE167x Computer Graphics (edX, free audit) · Scratchapixel — from-scratch graphics
■ CSCE 645 Geometric Modeling	<i>Curves, surfaces, solids: Bezier and spline theory, topology, Boolean operations, mesh processing.</i> · Keenan Crane, Discrete Differential Geometry (free notes + video + code) · libigl geometry processing tutorial (free) · MIT 6.8410 Shape Analysis — Justin Solomon

Course	What it is, and where to learn it free
■ CSCE 647 Image Synthesis	<i>Light transport from first principles: the rendering equation, Monte Carlo, importance sampling.</i> · Physically Based Rendering, 4th ed. — free online book · Ray Tracing in One Weekend series (free) · TU Wien Rendering / Light Transport course (free video)
CSCE 646 Digital Image	<i>2D image representation, filtering, warping, color spaces, and enhancement.</i> · Rich Radke, Digital Image Processing (RPI, free video) · OpenCV tutorials · scikit-image user guide
CSCE 648 Computer Aided Sculpting	<i>Form, proportion, and gesture expressed through parametric and implicit surfaces.</i> · Blender Fundamentals + sculpting (free, official) · Keenan Crane, Discrete Differential Geometry · Inigo Quilez — SDF and implicit modeling articles (free)
■ CSCE 649 Physically-Based Modeling	<i>Particles, rigid bodies, deformables, constraints, and fluids for animation.</i> · Baraff & Witkin, Physically Based Modeling (Pixar course notes, free) · Ten Minute Physics — Matthias Muller (free video + code) · Bridson, Fluid Simulation for Computer Graphics (course notes, free)
■ CSCE 650 Virtual Reality	<i>Perception, tracking, stereo rendering, latency, and the design of immersive experience.</i> · LaValle, Virtual Reality (free book + Illinois lectures) · Stanford EE267 — build your own HMD (free materials) · Unreal/Unity XR documentation
■ CSCE 748 Computational Photography	<i>Imaging pipelines, HDR, lightfields, coded apertures, and computational light transport.</i> · CMU 15-463 Computational Photography — Gkioulekas (free) · Marc Levoy, Lectures on Digital Photography (free) · Georgia Tech CS6475 Computational Photography (free)
CSCE 679 Data Visualization	<i>Visual encoding, perception, interaction, and narrative structure for data.</i> · Wilke, Fundamentals of Data Visualization (free book) · UW CSE 512 Data Visualization — Jeffrey Heer (free) · D3 / Observable Plot documentation
CSCE 743 Digital Fabrication Studio	<i>Machines, materials, and programmable matter as a computational design medium.</i> · MIT MAS.863 How to Make (Almost) Anything (free archive) · Fab Academy course archive · Prusa / open 3D printing knowledge base

Course	What it is, and where to learn it free
CSCE 653 Computer Methods in Applied Sciences	<i>Numerical analysis for engineers: roots, interpolation, quadrature, linear systems, ODEs/PDEs.</i> · MIT 18.330 Introduction to Numerical Analysis (OCW) · MIT 18.335 Numerical Methods — Steven Johnson (free notes) · Numerical Recipes / SciPy lecture notes (free)
■ CSCE 735 Parallel Computing	<i>Multicore, SIMD, GPU, and distributed-memory programming with a performance mindset.</i> · CMU 15-418 Parallel Computer Architecture & Programming (free) · Stanford CS149 (free slides + assignments) · NVIDIA CUDA C++ Programming Guide · OLCF CUDA Training Series (free video)
CSCE 654 Supercomputing	<i>Vectorization, numerical kernels, scaling studies, and performance measurement at HPC scale.</i> · HPC Carpentry (free lessons) · Argonne / ALCF training archive (free video) · CMU 15-418
CSCE 662 Distributed Processing Systems	<i>RPC, replication, consistency, fault tolerance, and distributed storage.</i> · MIT 6.5840 Distributed Systems (free video + labs) · van Steen & Tanenbaum, Distributed Systems (free book) · Google SRE Book (free)
CSCE 668 Distributed Algorithms and Systems	<i>Leader election, mutual exclusion, consensus, logical time, snapshots, fault tolerance.</i> · Suomela, Distributed Algorithms (free book) · MIT 6.852 Distributed Algorithms (OCW) · Raft / Paxos made live (free papers)
CSCE 678 Distributed Systems and Cloud Computing	<i>MapReduce, coordination, replication, elasticity, and the economics of the cloud.</i> · Cloud Computing Concepts — Indranil Gupta (free audit) · MIT 6.5840 · Google SRE Workbook (free)
CSCE 664 Wireless and Mobile Systems	<i>PHY/MAC fundamentals, scheduling, mobile transport, ad-hoc and sensor networks.</i> · Stanford CS144 (networking base) · NPTEL Wireless Communication (free video) · MIT 6.02 Intro to EECS II — digital communication (OCW)
CSCE 670 Information Storage and Retrieval	<i>Indexing, compression, ranking, and evaluation for very large document collections.</i> · Manning, Introduction to Information Retrieval (free book) · Search Engines: Information Retrieval in Practice (free PDF) · Stanford CS276

Course	What it is, and where to learn it free
CSCE 655 Human-Centered Computing	<i>Interaction concepts, prototyping, evaluation, visual design, and information structure.</i> · Klemmer, Human-Computer Interaction (free audit) · Nielsen Norman Group (free articles) · Stanford d.school resources (free)
CSCE 671 Computer-Human Interaction	<i>History, theory, task analysis, user modeling, and interaction styles.</i> · Klemmer, HCI (free audit) · Interaction Design Foundation open literature · ACM CHI proceedings (open access portion)
CSCE 672 Computer Supported Collaborative Work	<i>Technical systems for cooperative work, and their social and organizational consequences.</i> · ACM CSCW proceedings · Klemmer HCI (foundation) · MIT OCW — Collective Intelligence
CSCE 632 Accessible Computing	<i>Disability, universal design, accessibility standards, and inclusive software practice.</i> · W3C Web Accessibility Initiative tutorials (free) · WCAG 2.2 standard + Understanding docs (free) · Teach Access materials (free)
CSCE 656 Computers and New Media	<i>Computation as an authoring medium, and the author/reader relationship it creates.</i> · Lev Manovich, free books and essays · MIT OCW — Comparative Media Studies · Electronic Literature Collection (free)
CSCE 701 Foundations of Cybersecurity	<i>Crypto primitives, authentication, system and network defense, and the threat model habit.</i> · MIT 6.858 Computer Systems Security (OCW, free video) · Stanford CS155 Computer and Network Security (free) · RIT Cybersecurity Fundamentals (edX, free audit) · SEED Labs — free hands-on security labs
CSCE 711 Foundation of Modern Cryptography	<i>Provable security: PRGs/PRFs, block ciphers, MACs, public key, signatures, zero knowledge.</i> · Boneh & Shoup, A Graduate Course in Applied Cryptography (free) · Dan Boneh, Cryptography I (free audit) · Cryptopals challenges (free)
CSCE 749 Applied Cryptography	<i>MPC, verifiable computation, and zero-knowledge proofs applied to ML and blockchains.</i> · ZKP MOOC — Berkeley/Stanford (free video) · Evans, Kolesnikov & Rosulek, Pragmatic MPC (free book) · Boneh & Shoup (free)

Course	What it is, and where to learn it free
CSCE 713 Software Security	<i>Secure design, code review, static and dynamic analysis, fuzzing, and security testing.</i> · OWASP Top 10 + Cheat Sheet Series (free) · pwn.college — free hands-on exploitation curriculum · MIT 6.858 (free video)
CSCE 652 Software Reverse Engineering	<i>Binary formats, disassembly, control-flow recovery, static/dynamic analysis, obfuscation.</i> · OpenSecurityTraining2 — free RE courses · Ghidra (NSA) + official course materials (free) · LiveOverflow, Binary Exploitation (free video)
CSCE 712 Digital Forensic Engineering	<i>Acquisition, preservation, analysis, and the legal constraints on digital evidence.</i> · Autopsy / The Sleuth Kit documentation (free) · OpenSecurityTraining2 forensics tracks · NIST Computer Forensics Tool Testing (free)
CSCE 715 Secure Authentication Systems	<i>Passwords, MFA, biometrics, end-to-end protocols, and authentication side channels.</i> · NIST SP 800-63 Digital Identity Guidelines (free) · WebAuthn Guide (free) · FIDO Alliance specifications (free)
CSCE 716 Foundations and Applications of Blockchains	<i>Consensus and trust assumptions, smart contracts, and what blockchains actually buy you.</i> · Princeton, Bitcoin and Cryptocurrency Technologies (free book + video) · MIT 15.S12 Blockchain and Money — Gensler (OCW) · Ethereum developer documentation (free)
CSCE 765 Network Security	<i>Protocol vulnerabilities layer by layer, and the defenses that actually hold.</i> · Stanford CS155 (free) · SEED Labs network security labs (free) · Practical Networking — TLS/crypto series (free video)
CSCE 665 Advanced Networking and Security	<i>Offensive and defensive tooling against live networks and protocols.</i> · pwn.college (free) · SEED Labs (free) · Wireshark documentation + sample captures (free)
CSCE 702 Law and Policy in Cybersecurity	<i>Governance, oversight, procurement, and the regulatory landscape across jurisdictions.</i> · NIST Cybersecurity Framework 2.0 (free) · CISA free resources and guidance · Berkman Klein Center publications (free)

Course	What it is, and where to learn it free
CSCE 703 Cybersecurity Risk	<i>Risk identification, assessment, treatment, and audit frameworks.</i> · NIST SP 800-30 Risk Assessment (free) · NIST SP 800-37 Risk Management Framework (free) · FAIR risk quantification introduction (free)
CSCE 704 Data Analytics for Cybersecurity	<i>Clustering, supervised learning, anomaly detection, and visualization applied to attacks.</i> · MITRE ATT&CK (free) · Stanford CS229 (ML foundation) · Elastic Security / detection-rules (free, open)
CSCE 680 Testing and Diagnosis of Digital Systems	<i>Fault models, ATPG, scan chains, BIST, and design for testability.</i> · NPTEL VLSI Testing (free video) · Vishwani Agrawal, VLSI Testing lectures (free) · OpenLane / open EDA documentation
CSCE 681 Seminar	<i>Structured exposure to current research: read, present, and defend recent papers.</i> · SIGGRAPH / SIGGRAPH Asia open-access papers · arXiv cs.GR daily feed · Two Minute Papers (free video)
CSCE 643 Seminar in Intelligent Systems and Robotics	<i>Reading-group format tracking current robotics and intelligent-systems research.</i> · CoRL / ICRA / RSS open-access proceedings · arXiv cs.RO recent · Robot Brains / robotics seminar recordings
CSCE 689 Special Topics	<i>Instructor-chosen advanced topic; in this program, a self-designed deep dive.</i> · Open Syllabus — find a syllabus on any topic · MIT OpenCourseWare full catalog · Awesome-* curated topic lists on GitHub
CSCE 684 Professional Internship	<i>Supervised practice in a professional setting.</i> · Open-source contribution as substitute practice · Google Summer of Code · Outreachy
CSCE 685 Directed Studies	<i>A small, scoped research problem chosen to build technique.</i> · The Craft of Research (method, library access) · How to Read a Paper — S. Keshav (free) · Papers With Code

Course	What it is, and where to learn it free
CSCE 691 Research	<i>Sustained independent research toward a thesis.</i> · The Researcher's Bible (free) · How to Write a Great Research Paper — S. Peyton Jones (free) · arXiv + Zotero workflow
CSCE 682 Introduction to Doctoral Studies	<i>Orientation to doctoral milestones, publication, and presentation of work.</i> · PhD Grind — Philip Guo (free book) · How to Write a Great Research Paper (free) · The Researcher's Bible (free)